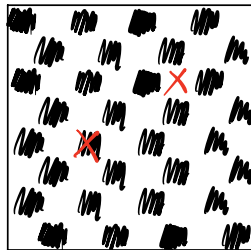


Is it possible to cover this chess board with dominos? NO!



Alice & Bob

$2n$ numbers

(10) (2) (100) (1) (5) (6) (3) (4)

A 10 2 3 6 = 21

B 4 100 1 5 = 110

Is there any strategy for Alice to get always at least as Bob?

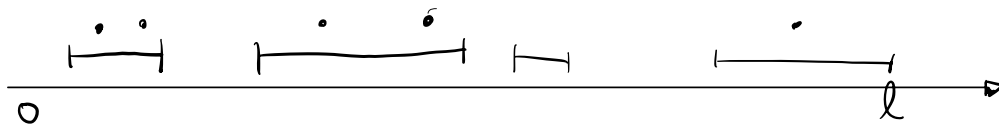
Binary Search for the answer

$$\sqrt{x}$$

Social Distancing

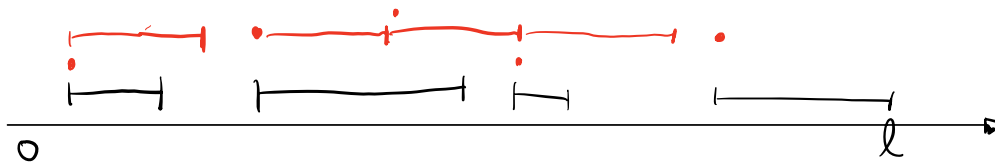


n disjoint intervals
 $c (= 5)$ people to place on these intervals



Goal: Find c points on the intervals such that the smallest distance between any of them is maximized

Target distance $d = 10$



PLACE POINTS WITH TARGET DISTANCE

$$\Theta(n+c)\text{time}$$

binary search for the answer $d^* \in [0, \frac{L}{c}]$
with $\log(\frac{L}{c})$ target distances to try in
 $\Theta(n+c)$ time each

FINAL COMPLEXITY is $\Theta((n+c) \log \frac{L}{c})$ time

EXPONENTIAL (BINARY) SEARCH

0 2^0 2^1 2^2 2^3 2^4 t 2^6 $\dots$

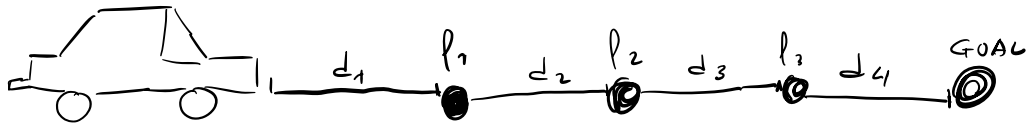


Phase #1 try powers of 2 till the first
true

$\Theta(\log(t))$ time

Phase #2 binary search in $[2^i \dots 2^{i+1}]$

$\Theta(\log(t))$ time



$$f = \text{fuel}/\text{km}$$

G^* min amount of fuel in our tank?

TWO-POINTERS TRICK

Problem

Longest k -good segment

Given $A[1, n]$ of values and a value k

A subarray is called **k -good** if it contains no more than k different values.

4 2 3 1 2 2 3 4
3-good

Find (a) longest one.

A 1 1 2 3 3 4 4 1 3 2 3

$k = 3$

H 1: 2 2: 1 3: 2 4: 3

$G = A[k]$

Time: $\Theta(n \log^* n)$ time

Sweep Line

Maximum Number of overlapping intervals

Given n intervals $[s_i, e_i]$ where $e_i, s_i \in \mathbb{N}$

we say that two intervals overlap

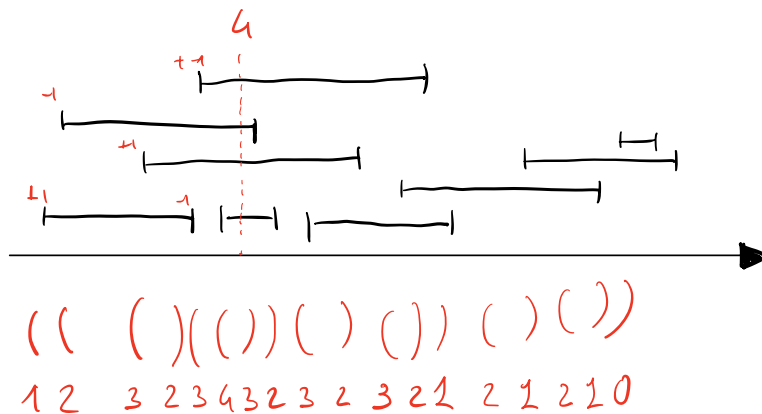
if $\exists x$ such that

$$x \in [s_i, e_i]$$

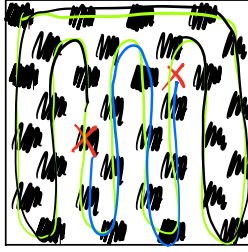
AND

$$x \in [s_j, e_j]$$

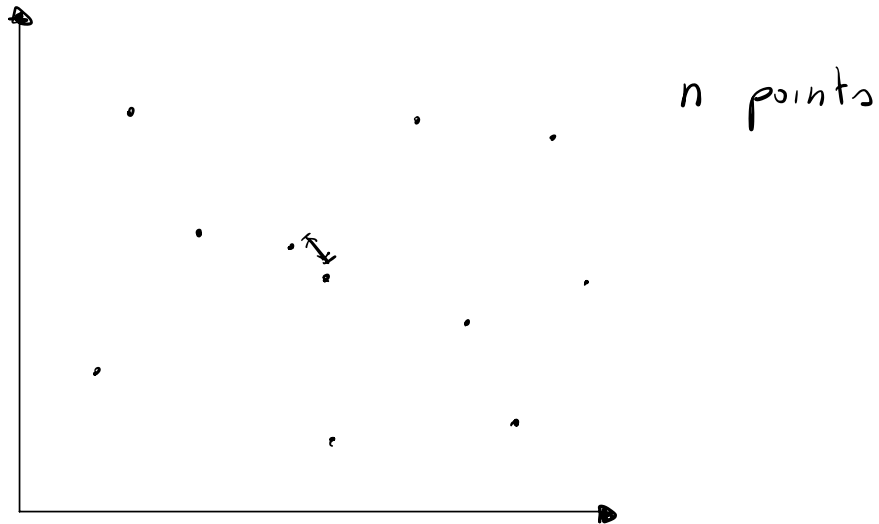
Compute the maximum number of overlapping intervals



Time complexity $\Theta(n \log n)$ time



Closest Pair of Points



Find the closest pair of points

Euclidean Distance

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Solution

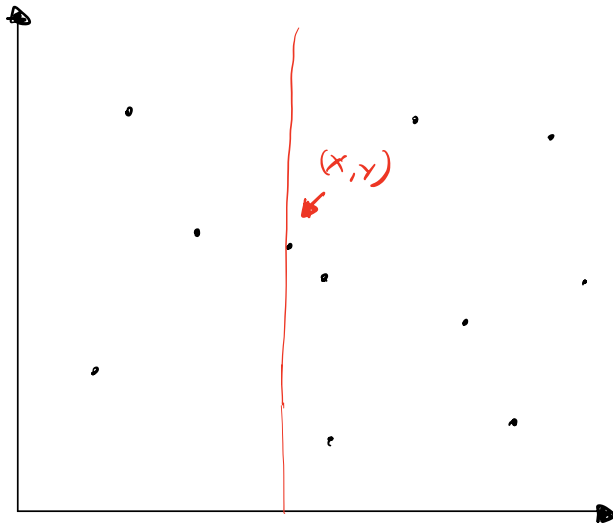
Brute-Force Solution

$\Theta(n^2)$ time

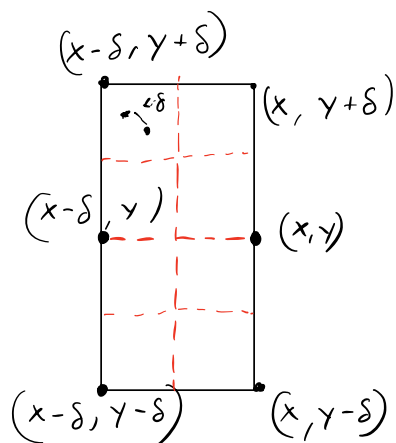
Target Solution

$\Theta(n \log n)$ time

δ = smallest distance
seen so far

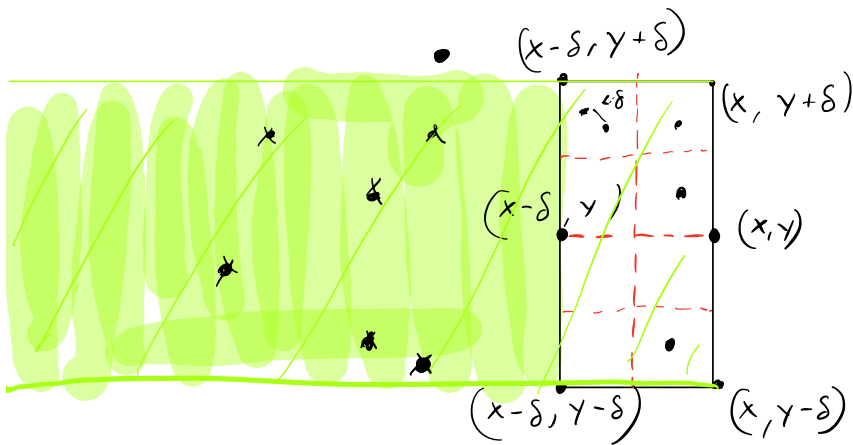


we know δ



It is possible to prove
that there are
at most 6 points
in the rectangle

- Store already processed points into a BST sorted by y

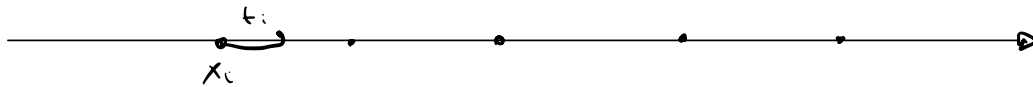


Time complexity : $\Theta(n \log n)$ time

Problem

Frogs and Mosquitoes

There are n frogs on the x -axis



Each frog (i) has a value t_i which is the length of its tongue

m mosquitoes start landing on x -axis

A mosquito j lands at position b_j and it has a value a_j .

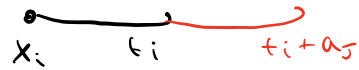
A frog i eats a mosquito b_j

if $b_j \in [x_i, x_i + t_i]$.

if there are many frogs able to eat b_j only the left most one eats it.



If frog x_i eats b_j with value a_j
then the new tongue is $t_i + a_j$



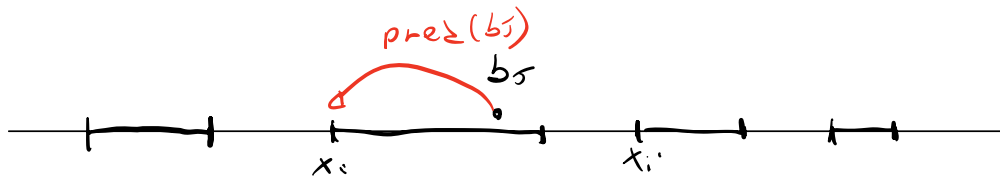
Notice that a mosquito b_j may be not eaten for a while



Target time complexity $\Theta((n+m) \log(n+m))$

Dynamic 1D point location

Solution



- Storing x_i in a BST together with t_i
- Predecessor of b_j is x_i which is the frog that eats the mosquito
- $\Theta(\log(n))$

• Issue #1



- Preprocess intervals to remove overlaps



- takes $\Theta(n \log n)$ time

- Issue #2: A interval is growing to right.
we are introducing new overlaps

repeat

- move to next prog

- remove it if now total overlap

- update its starting point

↑

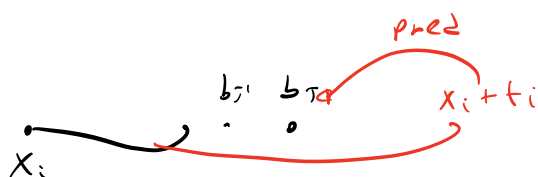
$\Theta(\log n)$ because just
for the last one

overlap cost
is $\Theta(n \log n)$

- Issue #3 not eaten mosquitoes

- Store mosquitoes in a BST

- Check if the last prog can eat a
old mosquito



$\Theta(m \log m)$